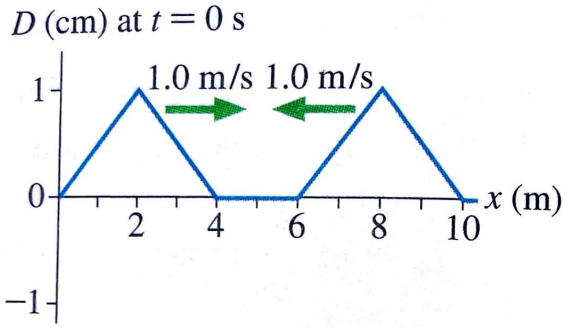
1. The figure is a snapshot graph at t = 0 s of two waves approaching each other at 1.0 m/s. Draw six snapshot graphs, stacked vertically, showing the string at 1 s intervals from t = 1 s to t= 6s.



2. Nylon Guitar strings have a linear density of 7.2 g/m and is under a tension of T = 150 N.

- **A.** Determine the wave speed in the string
- **B.** The string is plucked producing a wavelength of λ = 90 cm. Which frequency does this correspond to?

3. A 3 cm pulse travels down a string under a tension of 500 N. At a certain point a discontinuity is encountered where the linear mass density changes from 3 g/m to 5 g/m.

- **A.** Determine the wave speed in each region
- **B.** Determine the impedance in each region
- **C.** Determine the amplitude of the reflected wave
- **D.** Determine the amplitude of the transmitted wave
- **E.** Show that $A_I = A_T - A_R$.

4. Suppose a traveling Gaussian pulse has the following wave equation:

$$D(x,t) = Ae^{-(x-vt)^2}$$

- **A.** Use manipulate in mathematica to visualize the propagation of a pulse with an amplitude of A = 4, and a velocity of v =1 for the first 5 seconds.
- **B.** Create a second pulse that starts at x = 10 and travels backwards with the same amplitude as the first, but inverted
- **C.** (Extra Credit) Suppose the pulses are traveling on a string under a tension of 100 N. At x = 5, a rightward traveling pulse encounters a discontinuity where the linear mass density increases from 5 to 20 g/m. Use manipulate to visualize an incident wave encountering the discontinuity, and splitting into its transmitted and reflected modes.